

Multi-armed bandits and regret

Lecture 1 — Bandits & Regret, slides 1–63. Book chapter 2.

Definition 1. K -armed stochastic bandit

A K -armed stochastic bandit is a collection of K probability distributions $\nu_1, \dots, \nu_K$ over $[0, 1]$, with means $\mu(a) = \mathbb{E}_{r \sim \nu_a}[r]$. The learner interacts with it for a known horizon T : for $t = 0, 1, \dots, T - 1$,

1. the learner pulls an arm $A_t \in \{1, \dots, K\}$, chosen as a function of everything observed so far;
2. the environment draws $r_t \sim \nu_{A_t}$, independently of the past given A_t , and reveals r_t — and only r_t .

An optimal arm is any $a^* \in \arg \max_a \mu(a)$, and we write $\mu^* = \mu(a^*)$.

Definition 2. Regret

The regret of a policy over T rounds is

$$\text{Reg}(T) = T\mu^* - \sum_{t=0}^{T-1} \mu(A_t).$$

{#eq-regret}

Lemma 3. Regret decomposition

For every policy and every realisation,

$$\text{Reg}(T) = \sum_{a=1}^K \Delta(a) N_T(a).$$

Proof.

Group the sum in Lecture 1, slides 1–63 by the arm pulled:

$$\text{Reg}(T) = \sum_{t=0}^{T-1} (\mu^* - \mu(A_t)) = \sum_{t=0}^{T-1} \Delta(A_t) = \sum_{a=1}^K \Delta(a) \sum_{t=0}^{T-1} \mathbf{1}[A_t = a],$$

and the inner sum is $N_T(a)$ by definition.

Proposition 4. Greedy suffers linear regret

There is a two-armed bandit on which the greedy rule (one pull per arm, then commit to the larger empirical mean, ties to the lower index) has

$$\mathbb{E}[\text{Reg}(T)] \geq 0.032(T - 2) \quad \text{for every } T \geq 2.$$

Proof.

Take $\nu_1 = \text{Bernoulli}(0.6)$ and $\nu_2 = \text{Bernoulli}(0.4)$, so $a^* = 1$ and $\Delta(2) = 0.2$. After the two exploratory pulls, the rule commits to arm 2 exactly when it observed $r = 0$ from arm 1 and $r = 1$

from arm 2; in the other three outcomes it commits to arm 1, ties included. Those two events are independent, so

$$\mathbb{P}(\text{commit to arm 2}) = 0.4 \times 0.4 = 0.16.$$

On that event the remaining $T - 2$ rounds each pay $\Delta(2) = 0.2$, and by Lemma 3 the regret is at least $0.2(T - 2)$. Hence $\mathbb{E}[\text{Reg}(T)] \geq 0.16 \cdot 0.2 \cdot (T - 2) = 0.032(T - 2)$. **Algorithm A — Explore-then-commit (ETC)**

Input: horizon T , arms K , exploration budget m with $mK \leq T$.

1. **Explore.** For each arm $a = 1, \dots, K$: pull a exactly m times and record $\hat{\mu}(a) = \frac{1}{m} \sum_{i=1}^m r_{a,i}$.
2. **Commit.** Let $\hat{a} \in \arg \max_a \hat{\mu}(a)$. For the remaining $T - mK$ rounds, pull $\hat{a}$.

Theorem 5. Hoeffding's inequality

Let $X_1, \dots, X_n$ be i.i.d. random variables with values in $[0, 1]$ and mean μ , and let $\hat{\mu} = \frac{1}{n} \sum_{i=1}^n X_i$. Then for every $\varepsilon > 0$,

$$\mathbb{P}(|\hat{\mu} - \mu| \geq \varepsilon) \leq 2 \exp(-2n\varepsilon^2).$$

Corollary 6. Confidence radius

Under the hypotheses of Theorem 5, for every $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$|\hat{\mu} - \mu| \leq \text{rad}(n, \delta) = \sqrt{\frac{\ln(2/\delta)}{2n}}.$$

Proof.

Set $2 \exp(-2n\varepsilon^2) = \delta$ and solve for ε : the failure probability in Theorem 5 equals δ exactly at $\varepsilon = \sqrt{\ln(2/\delta)/(2n)}$.

Lemma 7. All arms at once

Run the exploration phase of Algorithm A. With probability at least $1 - \delta$,

$$|\hat{\mu}(a) - \mu(a)| \leq \sqrt{\frac{\ln(2K/\delta)}{2m}} \quad \text{for every arm } a.$$

Proof.

Apply Corollary 6 to each arm with failure probability δ/K : the event that arm a violates its interval has probability at most δ/K . The union of K such events has probability at most δ , and on its complement all K intervals hold. Substituting δ/K into $\text{rad}(m, \cdot)$ gives $\sqrt{\ln(2K/\delta)/(2m)}$.

Theorem 8. Regret of explore-then-commit

Run Algorithm A with $m \geq 1$ and $mK \leq T$. With probability at least $1 - \delta$,

$$\text{Reg}(T) \leq mK + T \sqrt{\frac{2 \ln(2K/\delta)}{m}}.$$

Proof.

Work on the event of Lemma 7, which has probability at least $1 - \delta$, and write $\varepsilon = \sqrt{\ln(2K/\delta)/(2m)}$ for the common radius.

Exploration. Each of the mK rounds has per-round regret at most 1, since rewards lie in $[0, 1]$ and hence every gap is at most 1. Its contribution is at most mK . (It is at most $m(K - 1)$, because m of those pulls are on an optimal arm, but the cruder bound is what the optimisation below uses.)

Commitment. The rest of the night pays the gap of the committed arm, so its contribution is $(T - mK) \Delta(\hat{a}) \leq T \Delta(\hat{a})$. Bound the gap: on the good event

$$\mu^* \leq \hat{\mu}(a^*) + \varepsilon \leq \hat{\mu}(\hat{a}) + \varepsilon \leq \mu(\hat{a}) + 2\varepsilon,$$

where the first and third steps are Lemma 7 applied to a^* and to $\hat{a}$, and the middle step is the definition of $\hat{a}$ as the maximiser of $\hat{\mu}$. Hence $\Delta(\hat{a}) = \mu^* - \mu(\hat{a}) \leq 2\varepsilon$.

The two phases together give $\text{Reg}(T) \leq mK + 2T\varepsilon = mK + T\sqrt{2\ln(2K/\delta)/m}$.

Corollary 9. The $T^{2/3}$ rate

Write $c = \sqrt{2\ln(2K/\delta)}$. The bound of Theorem 8 is minimised at

$$m^* = \left(\frac{Tc}{2K}\right)^{2/3},$$

and at that value, with probability at least $1 - \delta$,

$$\begin{aligned} \text{Reg}(T) &\leq 3 \cdot 2^{-1/3} T^{2/3} K^{1/3} \ln^{1/3}\left(\frac{2K}{\delta}\right) \\ &= O\left(T^{2/3} K^{1/3} \ln^{1/3}(2K/\delta)\right). \end{aligned}$$

Proof.

Let $f(m) = mK + Tcm^{-1/2}$ for $m > 0$. Then $f'(m) = K - \frac{1}{2}Tcm^{-3/2}$, which vanishes exactly when $m^{3/2} = Tc/(2K)$, i.e. at $m^* = (Tc/(2K))^{2/3}$; since $f''(m) = \frac{3}{4}Tcm^{-5/2} > 0$, f is convex on $m > 0$ and m^* is its minimum.

Write $u = (Tc/(2K))^{1/3}$, so that $m^* = u^2$ and $Tc = 2Ku^3$. Then

$$\begin{aligned} f(m^*) &= Ku^2 + \frac{Tc}{u} = Ku^2 + 2Ku^2 = 3Ku^2 \\ &= 3K \left(\frac{Tc}{2K}\right)^{2/3} = 3 \cdot 2^{-2/3} T^{2/3} K^{1/3} c^{2/3}. \end{aligned}$$

Substituting $c^{2/3} = 2^{1/3} \ln^{1/3}(2K/\delta)$ gives the claim, with $3 \cdot 2^{-2/3} \cdot 2^{1/3} = 3 \cdot 2^{-1/3} \approx 2.39$.

Lemma 10. Uniform confidence

Let $\hat{\mu}_{a,n}$ denote the empirical mean of the first n pulls of arm a . With probability at least $1 - \delta$, simultaneously for every arm $a \in \{1, \dots, K\}$ and every $n \in \{1, \dots, T\}$,

$$|\hat{\mu}_{a,n} - \mu(a)| \leq \sqrt{\frac{\ln(2KT/\delta)}{2n}}.$$

In particular the inequality holds at $n = N_t(a)$ for every round t with $N_t(a) \geq 1$.

Proof.

Fix a pair (a, n) . The first n rewards of arm a are i.i.d. draws from ν_a , since the algorithm chooses *when* to collect them and not what they are, so Corollary 6 applies with failure probability $\delta/(KT)$ and gives a radius $\sqrt{\ln(2KT/\delta)/(2n)}$. There are KT such pairs; by the union bound the probability that any of them fails is at most $KT \cdot \delta/(KT) = \delta$.

On the complement, the inequality holds for all (a, n) at once. Since $N_t(a)$ takes values in $\{0, 1, \dots, T\}$ and $\hat{\mu}_t(a) = \hat{\mu}_{a, N_t(a)}$ whenever $N_t(a) \geq 1$, the final claim follows: the bound was proved for *every* count, so it is true at the random one. **Algorithm B — UCB**

Input: horizon T , arms K , confidence parameter δ , multiplier β .

1. For $t = 0, \dots, K - 1$: pull each arm once.
2. For $t = K, \dots, T - 1$: pull

$$A_t \in \arg \max_a [\hat{\mu}_t(a) + b_t(a)], \quad b_t(a) = \beta \sqrt{\frac{\ln(2KT/\delta)}{N_t(a)}},$$

observe r_t and update N_{t+1} and $\hat{\mu}_{t+1}$.

Lemma 11. Per-round regret of an optimistic choice

On the event of Lemma 10, with $\beta \geq 1/\sqrt{2}$, for every round $t \geq K$,

$$\mu^* - \mu(A_t) \leq 2b_t(A_t).$$

Proof.

On that event $b_t(a)$ is at least the true radius, so for every arm $\mu(a) \leq \hat{\mu}_t(a) + b_t(a)$ and $\hat{\mu}_t(a) - b_t(a) \leq \mu(a)$. Then

$$\mu^* \leq \hat{\mu}_t(a^*) + b_t(a^*) \leq \hat{\mu}_t(A_t) + b_t(A_t) \leq \mu(A_t) + 2b_t(A_t),$$

where the middle inequality holds because A_t maximises $\hat{\mu}_t + b_t$.

Lemma 12. Sum of inverse square roots

For every integer $N \geq 1$, $\sum_{n=1}^N \frac{1}{\sqrt{n}} \leq 2\sqrt{N}$.

Proof.

The map $x \mapsto x^{-1/2}$ is decreasing, so for $n \geq 1$

$$\frac{1}{\sqrt{n}} \leq \int_{n-1}^n \frac{dx}{\sqrt{x}} = 2(\sqrt{n} - \sqrt{n-1}).$$

Summing over $n = 1, \dots, N$ telescopes to $2\sqrt{N}$.

Lemma 13. Concavity of the total

For non-negative integers $n_1, \dots, n_K$ with $\sum_a n_a = T$, $\sum_{a=1}^K \sqrt{n_a} \leq \sqrt{KT}$.

Proof.

By Cauchy–Schwarz applied to the vectors $(1, \dots, 1)$ and $(\sqrt{n_1}, \dots, \sqrt{n_K})$, $\sum_a \sqrt{n_a} \leq \sqrt{K} \sqrt{\sum_a n_a} = \sqrt{KT}$. (Equivalently, by Jensen’s inequality for the concave $\sqrt{\cdot}$: the sum is largest when the pulls are spread evenly, $n_a = T/K$.)

Theorem 14. Regret of UCB

Run Algorithm B with $\beta \geq 1/\sqrt{2}$. With probability at least $1 - \delta$,

$$\text{Reg}(T) \leq K + 4\beta \sqrt{KT \ln\left(\frac{2KT}{\delta}\right)} = \tilde{O}(\sqrt{KT}).$$

Proof.

Work on the event of Lemma 10. The first K rounds contribute at most K , one per round. For the remaining rounds, Lemma 11 and the definition of b_t give

$$\text{Reg}(T) \leq K + \sum_{t=K}^{T-1} 2b_t(A_t) = K + 2\beta \sqrt{\ln\left(\frac{2KT}{\delta}\right)} \sum_{t=K}^{T-1} \frac{1}{\sqrt{N_t(A_t)}}.$$

It remains to bound the sum, and here the trick is to stop following time and follow the arms instead. Group the rounds by the arm pulled: for a fixed arm a , the rounds $t \geq K$ at which $A_t = a$ see the counter $N_t(a)$ take the values $1, 2, \dots, N_T(a) - 1$, each exactly once, since the counter increases by one at each such round and equals 1 after the initialisation. Hence

$$\begin{aligned} \sum_{t=K}^{T-1} \frac{1}{\sqrt{N_t(A_t)}} &= \sum_{a=1}^K \sum_{n=1}^{N_T(a)-1} \frac{1}{\sqrt{n}} \leq \sum_{a=1}^K 2\sqrt{N_T(a)-1} \\ &\leq \sum_{a=1}^K 2\sqrt{N_T(a)} \leq 2\sqrt{KT}, \end{aligned}$$

where the inner sum is empty when $N_T(a) = 1$; the first inequality is Lemma 12 applied with $N = N_T(a) - 1$, the second is monotonicity of $\sqrt{\cdot}$, and the third is Lemma 13 with $\sum_a N_T(a) = T$. Substituting gives $\text{Reg}(T) \leq K + 4\beta \sqrt{KT \ln(2KT/\delta)}$.

Proposition 15. Optimism of the maximum

Let $\hat{\mu}(1), \dots, \hat{\mu}(K)$ be any unbiased estimates, $\mathbb{E}[\hat{\mu}(a)] = \mu(a)$, and fix a maximiser a^* of μ . Then

$$\mathbb{E}\left[\max_a \hat{\mu}(a)\right] \geq \max_a \mu(a) = \mu^*,$$

with equality if and only if $\max_a \hat{\mu}(a) = \hat{\mu}(a^*)$ almost surely: the largest estimate must belong to an arm that was already best.

Proof.

The function $(x_1, \dots, x_K) \mapsto \max_a x_a$ is convex, so by Jensen's inequality $\mathbb{E}[\max_a \hat{\mu}(a)] \geq \max_a \mathbb{E}[\hat{\mu}(a)]$, and the right-hand side is $\max_a \mu(a)$ by unbiasedness.

For the equality case, write $D = \max_a \hat{\mu}(a) - \hat{\mu}(a^*)$, which is non-negative pointwise. Since $\mathbb{E}[\hat{\mu}(a^*)] = \mu^*$,

$$\mathbb{E}\left[\max_a \hat{\mu}(a)\right] - \mu^* = \mathbb{E}[D],$$

and a non-negative variable has zero mean exactly when it is zero almost surely. So the bias is strictly positive unless the maximum is almost always taken at an arm that is optimal to begin with, which is what fails as soon as two arms are close and their estimates are noisy. The exercise measures the gap.

Example 16. Two booths, one token each

Two arms, both Bernoulli(1/2), so $\mu^* = 1/2$. Pull each once. Then $\max_a \hat{\mu}(a) = 1$ unless both rewards are 0, which happens with probability 1/4; hence

$$\mathbb{E}\left[\max_a \hat{\mu}(a)\right] = \frac{3}{4} \cdot 1 + \frac{1}{4} \cdot 0 = 0.75,$$

an overestimate of 50% built entirely out of true statements about honest tallies.